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Spin-related and non-spin-related effects of magnetic fields on water oxidation

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Water electrolysis is hindered by the slow kinetics and high overpotentials associated with the oxygen evolution reaction (OER), which takes place at the anode. Spin manipulation in the OER is a promising approach by which to modulate the reaction pathway to improve the energetics and kinetics. To that end, application of magnetic fields in the OER has been shown to enhance performance; however, whether the underlying promotional mechanisms are spin-related or non-spin-related remains a topic of ongoing debate. In this Review we explore OER enhancement under magnetic fields and elucidate both spin-related and non-spin-related effects, examining key fundamentals and experimental practices to distinguish these effects. For spin-related mechanisms, we highlight the key effects of spins on the catalyst bulk, catalytic interface and reaction intermediate. We provide guidance for understanding whether enhancements are spin-related or not.

In a transition to cleaner and more sustainable energy systems, hydrogen is a promising energy carrier. A key approach to producing clean hydrogen is electrolysing water using renewable electricity[1]. However, the overall reaction efficiency is impeded by the slow kinetics of the oxygen evolution reaction (OER), which occurs at the anode. Extensive studies have been reported for improving the intrinsic activity of the OER and understanding the underlying fundamentals[2].

The OER consists of four electron-transfer steps with intermediates such as *OOH, *O and *OH. The adsorption energies of these oxygenated intermediates are linearly correlated[3,4]. This interdependence of adsorption energies prevents the full optimization of the electrocatalysts, resulting in thermodynamic limitations. Various approaches have been used to overcome these so-called scaling relationships through catalyst design[2]. One emerging idea to address this challenge in the OER is spin-related electrocatalysis. For example, by optimizing spin-related interactions between the active site and intermediates, the direct coupling of neighbouring *O species—critical intermediates in the OER—with parallel spin alignment can be facilitated. This potentially bypasses the energy scaling between key oxygenated intermediates (for example, the Gibbs free energy difference $\Delta G(\text{*OOH}) - \Delta G(\text{*OH})$ fix at around 3.2 eV (ref. 3) to lead to a high theoretical overpotential), and could lead to OER pathways with more favourable energetics and kinetics.

It has been proposed that spin-related mechanisms are essential in the OER. This arises from the fact that the conversion of H_2O/OH^- to O_2 involves a singlet-to-triplet transition, and in quantum mechanics, any process that involves a change of spin typically has an additional enthalpy barrier[5], for which additional mechanisms to preserve the spin conservation during catalytic conversions have been proposed in the OER. Extensive research efforts have been made to study the effect of spins in OER kinetics[6]. In 2015, the effect of electron spin on the OER was explored via the chiral-induced spin selectivity (or CISS) effect[7], where electrons transported through a chiral-molecule-coated electrode become spin-polarized, and such a polarization was believed to promote the OER. Theoretical studies then demonstrated that favourable OER kinetics should be mediated by quantum spin-exchange interactions (QSEIs), which are more likely to occur on the surface of catalysts with ferromagnetic (FM) spin ordering[8]. A groundbreaking study then reported an enhancement in the OER by applying an external magnetic field to an electrolysis cell[9]. Since several magnetic ferrites (for example, $NiZnFeO_x$ and $NiZnFe_4O_x$) were reported[9], magnetic field-enhanced OER has been disclosed for various materials with FM spin ordering, including magnetic spinel oxides (for example, $CoFe_2O_4$ (ref. 10), Fe_3O_4@$Ni(OH)_2$ (ref. 11), $Co_{3-x}Fe_xO_4/Co(Fe)O_xH_y$ (ref. 12) and $ZnFe_2O_x$ (ref. 9)) and magnetic alloys (for example, $SmCo_5$ (ref. 13), FeNi (ref. 14) and $Ni_{0.8}Fe_{0.15}Mo_{0.05}$ (ref. 15)).

Nevertheless, challenges persist in the study of the OER under magnetic fields because complicated effects coexist, and thus it would be prudent not to attribute the OER enhancement solely to the spin-related effects. Non-spin-related effects, which include

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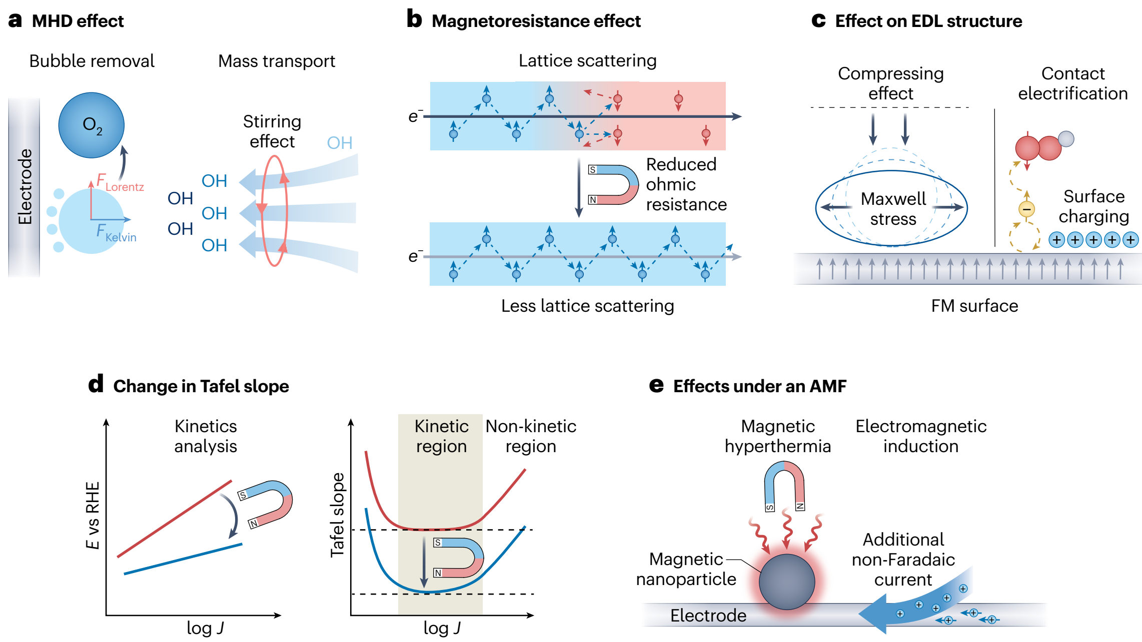

Fig. 1 | Non-spin-related effects generated in the electrochemical reaction under magnetic fields. **a**, Schematic illustrations of the MHD effect under magnetic fields, which include improved mass transport and bubble detachment effected by Lorentz ($F_{Lorentz}$) and Kelvin (F_{Kelvin}) forces. **b**, Schematic illustrations of the magnetoresistance effect in FM materials, originating from the lattice scattering of spin during electron transport. The blue and red regions represent, respectively, the spin-up and spin-down electrons (e^-) in FM materials. **c**, Influence on the electrical double layer (EDL) structure under magnetic fields. The EDL can be affected via the compressing effect of Maxwell stress and spontaneous surface charging through contact electrification. The blue-coloured solid and dashed oblates on the left represent the deformation of the paramagnetic phase by Maxwell stress. The yellow-coloured spin on the right represents an electron from the FM material, which forms a radical pair [e^-••HO_2] with the protonated O_2 (HO_2). **d**, The change in Tafel slope due to kinetic and non-kinetic effects. The magnetic enhancement should be evaluated from the kinetically meaningful Tafel slopes (brown region). E, electrode potential; RHE, reversible hydrogen electrode; J, current density. **e**, Schematic illustrations of magnetic hyperthermia and the electromagnetic induction effect under an alternating magnetic field (AMF). Panels adapted with permission from: **c**, left, ref. 19, American Chemical Society, and right, ref. 35 under a Creative Commons licence CC BY 4.0; **e**, ref. 18, Springer Nature Limited.

but are not limited to the magnetohydrodynamic (MHD) effect[16], magnetoresistance[17], magnetic hyperthermia[18], Maxwell stress[19], among others, are also reported to boost OER activity. To investigate the spin-related OER enhancement by magnetization, it is essential to understand these complicated effects under magnetic fields as well as the origin of the spin-related magnetization-induced enhancement in the OER.

In this Review we summarize the spin-related and non-spin-related effects of magnetic fields on the reported OER enhancement, providing insights into the underlying fundamentals and recommending key experimental practices to distinguish these effects. We highlight the roles of spin-related effects in the catalyst bulk, at the catalytic interface and on reaction intermediates, and we discuss the related prerequisites for observing spin-related enhancement. Finally, we provide an outlook for electrocatalysis under magnetic fields beyond achieving just an improvement in activity, underscoring the need to uncover more spin physics that are associated with the inherent limitations in the OER.

Non-spin-related effects under a magnetic field

Applying an external magnetic field to an electrochemical system increases the complexity of the measurement. Multiple phenomena coexist alongside the spin-related effects, complicating the identification of the main origin of the OER enhancement by magnetization. This section briefly reviews the complicated effects associated with OER measurements under external fields (Fig. 1).

First, the MHD effect is a widely studied phenomenon that occurs when an electrochemical process is carried out under a magnetic field, and it is known to facilitate bubble removal and mass transport during reactions. The process involves two dominant forces: the Lorentz force and the Kelvin force. In a static magnetic field, the Lorentz force ($F_{Lorentz} = J \times B$, where J is the current density and B is the magnetic field) induces ion movement, thus resulting in a stirring effect in the electrolyte[20]. In the presence of FM materials, the magnetic field is redistributed, forming a magnetic field gradient that generates the Kelvin force[21].

The MHD effect can improve mass transport and the reaction performance in many mass-transport-limited reactions[20]. However, its impact on the OER enhancement under magnetic fields remains inconclusive. In strong alkaline electrolytes (for example, 1 M KOH) at low current densities, the mass transport of OH^- may affect the OER only limitedly. However, when the OH^- supply becomes rate-limiting, such as at elevated current densities or with weak alkaline electrolytes, the MHD effect has been found to improve the OER notably[22]. Therefore, it is crucial to distinguish the nature of the enhancement depending on the conditions under which the OER is carried out (for example, the current density and reactant concentration). Besides, it has been proposed that the influence of the MHD effect can depend on the transport mode of OH^- in bulk electrolytes[23]. Vehicular mode refers to a physical movement of OH^-, which causes a significant stirring effect under the Lorentz force, whereas the Grotthuss hopping mode describes the transport of OH^- in electrolytes via proton hopping without the physical movement of charged ions[24], which can minimize the stirring effect under magnetic fields.

Under a magnetic field oriented in a certain direction, bubble detachment can also be facilitated by both macro- and micro-scale

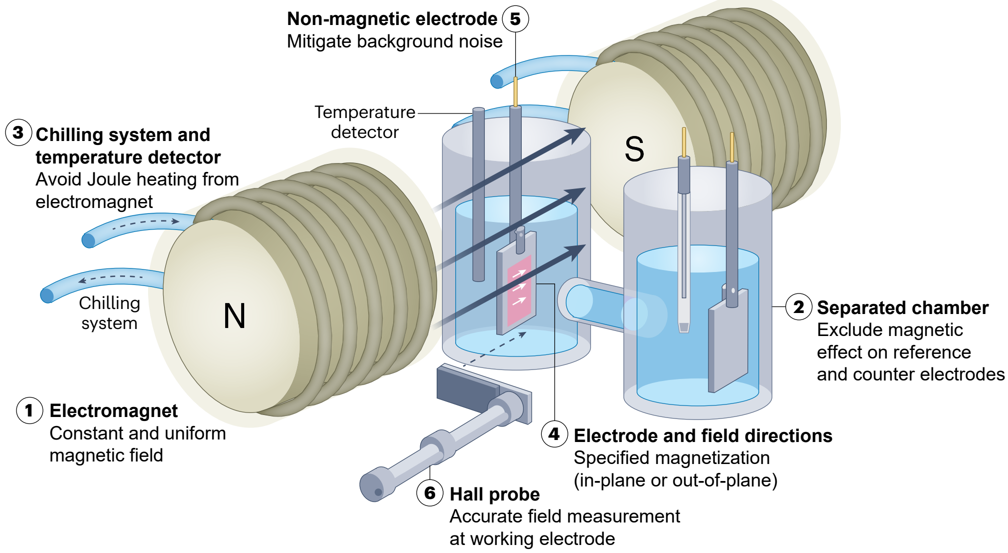

Fig. 2 | Recommended experimental set-up for OER measurement under a magnetic field. This design intends to identify the spin-related OER enhancement from complex effects under magnetic fields, where other effects are mitigated according to the notes beside the illustrations. A similar set-up has been demonstrated in key studies of the OER under magnetic fields, yielding reproducible and reliable results[20]. The set-up under an electromagnet is adapted with permission from ref. 24, Elsevier, and ref. 20 under a Creative Commons licence CC BY 4.0.

MHD (Fig. 1a), which shows an improvement in OER activity by reducing the ohmic voltage drop on the electrode surface and exposing more active sites[16]. The maximum macro-scale MHD occurs when the field is parallel to the electrode surface ($J \perp B$). An upward F_{Lorentz} is believed to facilitate O_2 bubble release. Conversely, when the field is perpendicular to the electrode ($J \| B$), a micro-scale MHD vortex can also form from the distorted current line at surface bulges to induce bubble removal[25]. However, how the bubbles are removed from electrodes is complicated, and a fundamental understanding of the bubble-removal mechanism should be accompanied by an understanding of the bubble dynamics on the electrode surface[26]. The detachment period and bubble sizes can be influenced by different forces such as the buoyancy force, electrical force and thermal and solutal Marangoni forces[26], which can be sensitive to the MHD-induced ion movement. The overall extent of the MHD effect on OER enhancement can depend on the current densities, the strength and direction of magnetic fields and the experimental set-ups[20,21,25,27]. In some cases, the related OER enhancement can exceed 60% (ref. 28).

Second, the magnetoresistance is an important variable under a magnetic field, especially for FM materials such as multilayer magnetic superlattices[29], $MoS_2/CoFe_2O_4$ heterostructures[30] and $CoFe_2O_4$ nanoparticles[31]. This effect, which originates from electron–electron scattering, alters the intrinsic resistance of FM materials[32]. In FM materials, the momentum scattering differs for spin-up (↑) and spin-down (↓) electrons, leading to distinct conductivity paths (Fig. 1b). The conductivity can be assumed to be the parallel connection of the $R_{\uparrow}$ and $R_{\downarrow}$ channels ($R_{\uparrow}$ and $R_{\downarrow}$ represent the resistance by spin-up and spin-down electrons, respectively), and parallel spins lead to minimal scattering of electrons[33]. Magnetization thus enhances the conductivity of FM materials by aligning spin channels.

Magnetoresistance may not be the main cause of the OER enhancement by magnetization in many cases. First, magnetoresistance is highly dependent on temperature due to thermal fluctuations, being negligible at room temperature compared to low temperatures[30,34]. Second, catalysts are typically mixed with conductive carbon to ensure high conductivity, where magnetoresistance variation contributes limitedly to the activity. Third, *iR* correction (*i*, current; *R*, resistance) is often performed to determine the real potential applied to catalysts, which can also correct the influence of magnetoresistance on the electronic resistance.

Third, the thickness and charge distribution of the electrical double layer (EDL) is critical for the OER kinetics. Some studies have shown the related effects of magnetic fields on the EDL structure during the OER, including Maxwell stress[16,19] and contact electrification[35]. Maxwell stress describes the deformation of the paramagnetic (PM) liquid phase under a magnetic field to change the wetting between the liquid and the electrode. It is reported that an in-plane magnetic field on an electrode can cause an oblate-like deformation in the nitrobenzene radical anion cloud during the nitrobenzene redox reaction, bringing the average outer Helmholtz plane closer to the electrode surface by up to 0.25 nm (Fig. 1c)[19]. In the OER, many electrocatalysts are mesoporous materials[36], where the double layer will be confined within the pores, enabling limited space for a diffuse layer. Compressing the double layer can bring the ions closer to the surface, which is likely to facilitate the OER on mesoporous catalysts[37]. Moreover, contact electrification between an O_2-containing liquid and an FM surface can be influenced by magnetic fields[35]. At the liquid–solid contact interface, electron transfer occurs between the dissolved PM O_2 molecule and the electrode surface due to the differences in the Fermi levels[35]. In the case of the FM Fe_3O_4 surface, electron transfer is facilitated by the external magnetic field via the Zeeman interaction, as shown by the increased surface charge under a 0.5 T magnetic field. The enhanced contact electrification is explained by the radical pair mechanism[35], where the protonated O_2 (that is, HO_2) and a $3d^6$ electron from Fe_3O_4 form a radical pair $[HO_2{\bullet\bullet}e^-]$. Electron transfer can be facilitated when the two spins are antiparallel (Fig. 1c). The triplet–singlet spin conversion of the radical pair can be triggered under a magnetic field, increasing the electron transfer at the interface. Such spontaneous charge transfer leads to additional charge accumulation on the electrode surface under a magnetic field, which potentially affects the EDL during the OER.

Fourth, the Tafel slope has been widely used to understand OER kinetics[38]. For FM materials, OER enhancement under a magnetic field is typically accompanied by a decrease in the Tafel slope (Fig. 1d, left)[10,39,40]. On the other hand, PM materials typically exhibit an unchanged Tafel slope under a magnetic field, especially at lower overpotentials[9,10]. However, an interesting phenomenon has been reported in PM CoO_x, where the magnetic enhancement consisted of two distinct linear regions with different field intensities[41]. For B_{anode} < 40 mT, the change in Tafel slope with the magnetic field is calculated to be ~135.3 μV dec^{-1} mT^{-1}, whereas the value decreases to ~22.03 μV dec^{-1} mT^{-1} for B_{anode} > 40 mT. However, one should be cautious about whether the revealed change in the Tafel slope is kinetically meaningful or not. In the OER, non-kinetic effects (for example, bubble accumulation and mass-transport limitation)[36] can be notable at higher current densities, which can be significantly influenced by the MHD effect. It is thus recommended to plot the Tafel slope as a function of the OER current density (Fig. 1d, right)[42], which could help to identify whether the magnetic effect is kinetic or not.

Finally, apart from the static magnetic field, an alternating magnetic field (AMF) can also affect the OER activity through complicated magnetic heating effects. Typically, under an AMF, magnetic hyperthermia for some FM materials, involving Néel and Brownian relaxation[43], will cause local magnetic heating to enhance the OER without heating the bulk electrolyte (Fig. 1e)[18]. Local magnetic heating under an AMF during the OER has been reported in FeC–Ni nanoparticles[18], NiCoFe-MOF-74 (ref. 44), NiFe/NiFeOOH (ref. 45) and pyrolysed ZIF-67 (ref. 46). However, one should be cautious here because significant heating effects due to marked eddy currents caused by electromagnetic induction in many conductive electrode materials[47] can significantly increase the temperature of the bulk electrolyte. In addition, electromagnetic induction can influence the electrode voltage and surface charge. For example, an enhanced non-Faradaic current has been reported in the OER under increasing AMF intensities[44].

Recommended experimental set-up

Most of the above-mentioned phenomena can be avoided using specific electrochemical set-ups, such as that illustrated in Fig. 2, and by carrying out necessary experimental practices. The use of an electromagnet, rather than a permanent magnet, is suggested as it generates a more uniform and strength-controllable magnetic field. A uniform and accurate magnetic field on the position of the working electrode, which could be measured using a Hall probe, is essential for studying the quantitative correlation between the spin polarization and OER activity. The field gradients are typically omitted in the set-ups with uniform magnetic fields. However, the effect of field gradients is an important and independent topic[21,48] that requires additional insights and experimental validation for a better understanding of the OER enhancement under magnetic fields. In addition, monitoring the temperature of the cell within the electromagnet is important for evaluating the effect of Joule heating from the coil of the electromagnet on the OER, especially with prolonged operation. It is critical to maintain the minimized change in temperature during measurement under a magnetic field.

Furthermore, in an H-type electrochemical cell with a three-electrode configuration, it is recommended to place only the working electrode compartment in the centre between the electromagnet tips to minimize interference from the counter and reference electrodes. A large working electrode can help to minimize electromagnetic induction by turning the magnetic field on and off[49]. When testing electrocatalysts, magnetic substrates such as Ni foam should be avoided, to exclude the contribution from substrate magnetization and the effects of electrode displacement (such as bending, tilting and vibrating) under a magnetic field that are likely to change the ohmic resistance in electrochemical reactions[41]. Throughout the experiment, ***iR*** correction should be performed to calibrate the potential of the working electrode[42].

Moreover, applying the field direction correctly to the working electrode is crucial. For materials with magnetic anisotropy, such as magnetic thin films, the magnetic field should follow the easy axis, which is the favourable direction of spontaneous magnetization[50]. On the other hand, for materials without magnetic anisotropy, such as powder catalysts, it is recommended that the magnetic field is applied parallel to the current flow near the working electrode, to minimize the macroscopic MHD effect[25].

Spin-related mechanisms in the OER

The spin-related mechanism in OER kinetics is widely believed to be closely associated with the observed OER enhancement by magnetization. However, the existing understanding of spin polarization under magnetic fields may not fully account for the actual spin-related reaction pathways. For example, it would be challenging for the magnetic field strengths (often below 1 T) that are typically applied during electrochemical reactions to induce direct spin flipping in the oxygen intermediates[51]. Merely conceiving a direct spin flipping at the atomic level under a magnetic field provides an insufficient understanding of spin-related effects.

Accordingly, important advancements have been made in understanding the spin-related effects associated with specific electrocatalysts. For example, spin-related effects have been suggested to play a role in the catalyst bulk, at the catalyst interface and in reaction intermediates (Fig. 3).

In 2019, direct OER enhancement was reported in magnetic $NiZnFeO_x$ by applying a permanent magnet near the working electrode to achieve long-range spin ordering in catalysts[9]. Since then, an increasing number of studies have validated enhancement of the OER by magnetization, attributing it primarily to spin polarization. Figure 4a summarizes those findings, illustrating a strong association between the enhancement and the long-range magnetic ordering of magnetized electrocatalysts. Notably, for most FM electrocatalysts, the pronounced OER improvement occurred with a magnetic field strength higher than the corresponding saturation field strength (H_S). Attention has been paid to the noticeable OER enhancement in superparamagnetic (SPM) catalysts, which achieve spin polarization with nearly zero H_S (ref. 52). Conversely, no observable OER enhancement under an external magnetic field was found for antiferromagnetic (AFM) or PM materials such as $LiCoVO_4$ (ref. 53) and IrO_2 (ref. 10), respectively. Therefore, it is generally believed that the spin polarization, as well as achieving long-range magnetic ordering in electrocatalysts, promotes the OER[10].

Figure 4a reveals a large disparity in the degree of activity improvement among FM materials. This disparity can be seen in the different reports even for FM electrocatalysts with identical chemical compositions[10,39]. The answer to this disparity calls for a fundamental understanding of the physics behind spin polarization.

Factors that contribute to spin polarization in FM materials may include the saturation magnetization (M_S), coercivity (H_C) and domain structure, correlating to the enhanced OER activities in previous studies. For example, the M_S (Fig. 4b) and H_C (Fig. 4c), which are two intrinsic properties of FM materials, positively correlate with the increased OER activity under magnetic fields by tuning the chemical composition or doping in the same catalyst systems. In addition, the long-range spin ordering in materials has also been demonstrated to depend on the magnetic domain structure. FM materials typically involve separated magnetic domains to minimize the magnetostatic energy[54], with highly ordered spins within domain regions and disordered spins at domain walls. Magnetization induces a transition from multi-domain to single-domain structures, in which the domain walls disappear (Fig. 4d). In 2023, a straightforward correlation was found between the degree of OER enhancement and the domain-wall ratio in NiFe thin films[14], indicating that the proportion of domain walls before magnetization governs the degree of OER enhancement by magnetization (Fig. 4e). It is concluded that the origin of the OER increment under a magnetic field lies in the reformatted domain structure and disappeared domain walls in FM catalysts. Therefore, the different magnetic properties of M_S, H_C and the domain walls provide a potential physical model by which to understand the disparity in OER enhancements, spin polarization in the bulk of FM electrocatalysts and the correlated OER activities. In addition to the intrinsic magnetic properties of electrocatalysts, it is important to note that most of the reported electrocatalysts are oxides, which are non-stoichiometric materials. These materials,

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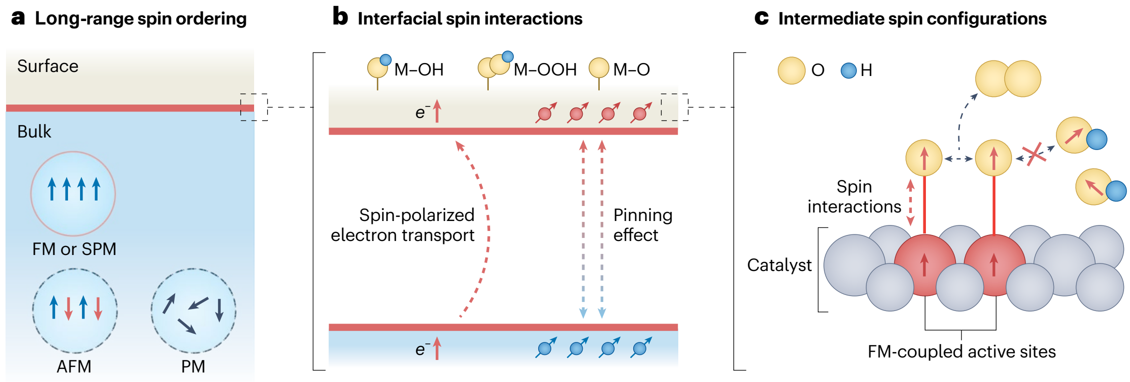

Fig. 3 | Effects specific to bulk, interfacial and intermediate spins. **a**, Long-range spin ordering in FM and SPM materials, showing ordered spin alignment, whereas the spin direction is antiparallel in AFM materials and disordered in PM materials. **b**, Interfacial spin interactions as a consequence of spin-polarized electron transport and the interfacial pinning effect. M, metal. **c**, Spin configurations in reaction intermediates to optimize the reaction energetics and kinetics.

even with same nominal composition, can differ in stoichiometry, surface morphology and crystalline faces, and often undergo surface reconstruction under operando conditions. Consequently, the OER enhancements under magnetic fields can vary significantly even for catalysts that share the same nominal composition[10,39].

Whereas enhancement is typically associated with FM catalysts, in alkaline electrolytes, many OER-active catalysts, such as state-of-the-art (oxy)hydroxides[12,17] and perovskites[55], are PM or AFM, for which long-range FM ordering is difficult to achieve. Moreover, the alkaline OER typically enables surface reconstruction on some electrocatalysts, leading to a surface dominated by oxyhydroxide chemistry[56]. Accordingly, spin manipulation in AFM and PM species is challenging but highly desirable for improving the alkaline OER[12]. Innovative approaches involving interface engineering to modify the interfacial magnetism, which affects the spin ordering in the reconstruction layer, have been proposed (Fig. 5a)[12,13].

For a typical FM–AFM(PM) interface in OER electrocatalysts, the spins near the interface will be aligned by an interfacial spin pinning effect (Fig. 5b)[57]. The pinned spins are 'frozen' at the interface, also known as the spin-glass state, which results in the exchange bias field (H_E) at the interface[58], and is difficult to be changed. The spin pinning effect has been widely reported in interfaces derived by surface reconstruction[12,13] and core–shell structures[11,59]. Notably, spin pinning is effective within a limited length, which calls for interface engineering with controlled surface layer thicknesses (typically <5 nm)[60]. A similar effect on interfacial spin alignment has been proposed as a type of Ruderman–Kittel–Kasuya–Yosida (or RKKY) interaction[61], which highlights the interplay between conduction electrons and localized magnetic moments in the non-magnetic materials. For example, the room-temperature PM Cr_2Te_3 nanosheet exhibits long-range FM magnetic ordering when in contact with an AFM CrOOH surface[62]. The induced FM character in the Cr_2Te_3/CrOOH heterostructure can promote the OER (ΔE = 70 mV at 10 mA cm^{-2}) after 0.7 T magnetization.

In addition, the built-in electric field generated at the interface of p-type and n-type materials affects the surface magnetization[63]. Within the p–n junction, electrons from the n-type region diffuse into the p-type region and recombine with holes, creating a depletion region with limited mobile charge carriers. A spin-polarized p–n junction was reported by growing NiFe-LDH/Co_3O_4 on Ni foam[63]. Spin injection from the Ni foam under a magnetic field may result in a non-equilibrium spin accumulation in the n-type NiFe-LDH, facilitating spin-polarized electron transport in the depletion layer (Fig. 5c)[64,65]. Evidence of this effect was shown experimentally by the enhanced OER activity (ΔE = 25 mV at 50 mA cm^{-2}) of the NiFe-LDH-Co_3O_4/Ni foam catalyst under a magnetic field[63].

Moreover, the double-exchange interaction is widely used to explain the FM ordering in mixed-valence compounds through the electron-hopping process between adjacent metal ions[66]. Such an interaction has been found to influence spin transport in the core–shell Co_3O_4/NiFeOOH electrocatalyst during the OER[66]. Electron hopping via Co_{Td}^{2+}–O^{2-}–Ni_{Oh}^{3+} and Co_{Td}^{2+}–O^{2-}–Fe_{Oh}^{3+} pathways, where the subscripts Td and Oh denote tetrahedral and octahedral sites, respectively, has been demonstrated to be spin-dependent, where the FM-coupled metal ions create polarized spin-conduction channels at the interface (Fig. 5d). Consequently, the Co_3O_4/NiFeOOH electrocatalyst, which has a strong double-exchange interaction, shows an OER activity that is ~26-fold higher than that of the NiFeOOH catalyst[66].

The interactions at the interface provide insights into spin polarization from the bulk to the surface of electrocatalysts, and broaden the spin-related promotion for more electrocatalysts without intrinsic FM spin ordering.

A comprehensive understanding of spin-related effects also demands insight into the atomic-level spin-related interactions in OER kinetics, which promotes O_2 turnover in a triplet state from OER intermediates. Theories have been proposed according to the existing experimental findings, highlighting the fundamental understanding of open-shell QSEIs[8,67], intermediate spin polarization[68] and spin selectivity[6,12] in OER kinetics (Fig. 6).

FM exchange interactions in magnetic oxides can optimize the binding energy of the reactant via open-shell QSEIs (Fig. 6a)[8,67], which primarily explain the higher OER activity in catalysts with FM coupling than in those with AFM coupling. The exchange interactions between two electrons on orbitals φ_a and φ_b are approximated from the Heisenberg exchange Hamiltonian $\mathcal{H} = -J_{ex}(\hat{s}_a\hat{s}_b)$, where the spin momenta of the electrons are given as $\hat{s}_a$ and $\hat{s}_b$, respectively, and J_{ex} is the exchange integral between the two spins[8]. If J_{ex} is positive, the exchange energy favours parallel spin, forming the FM exchange interaction; conversely, a negative J_{ex} results in an AFM exchange interaction. Once the intra-atomic QSEIs stabilize the open-shell configuration, the interatomic FM exchange interaction facilitates coherent electron-spin transport. This enhancement is favoured by the reduced Coulomb repulsion between two spin-parallel electrons in different orbitals when the two electrons are approaching and interchanging their orbitals and energy momentum (Fig. 6b)[69]. In 3*d*, 4*d* and 5*d* metal oxides, the interatomic exchange interaction is typically FM and dominates over the crystal field energy, Jahn–Teller distortion or spin–orbit couplings. Accordingly, the reduced Coulomb repulsions between electrons enhance the electron-spin transport during the reaction, facilitated by the QSEIs. The effect on the OER binding energy can be described by the electron-hopping process in the magnetic mixed

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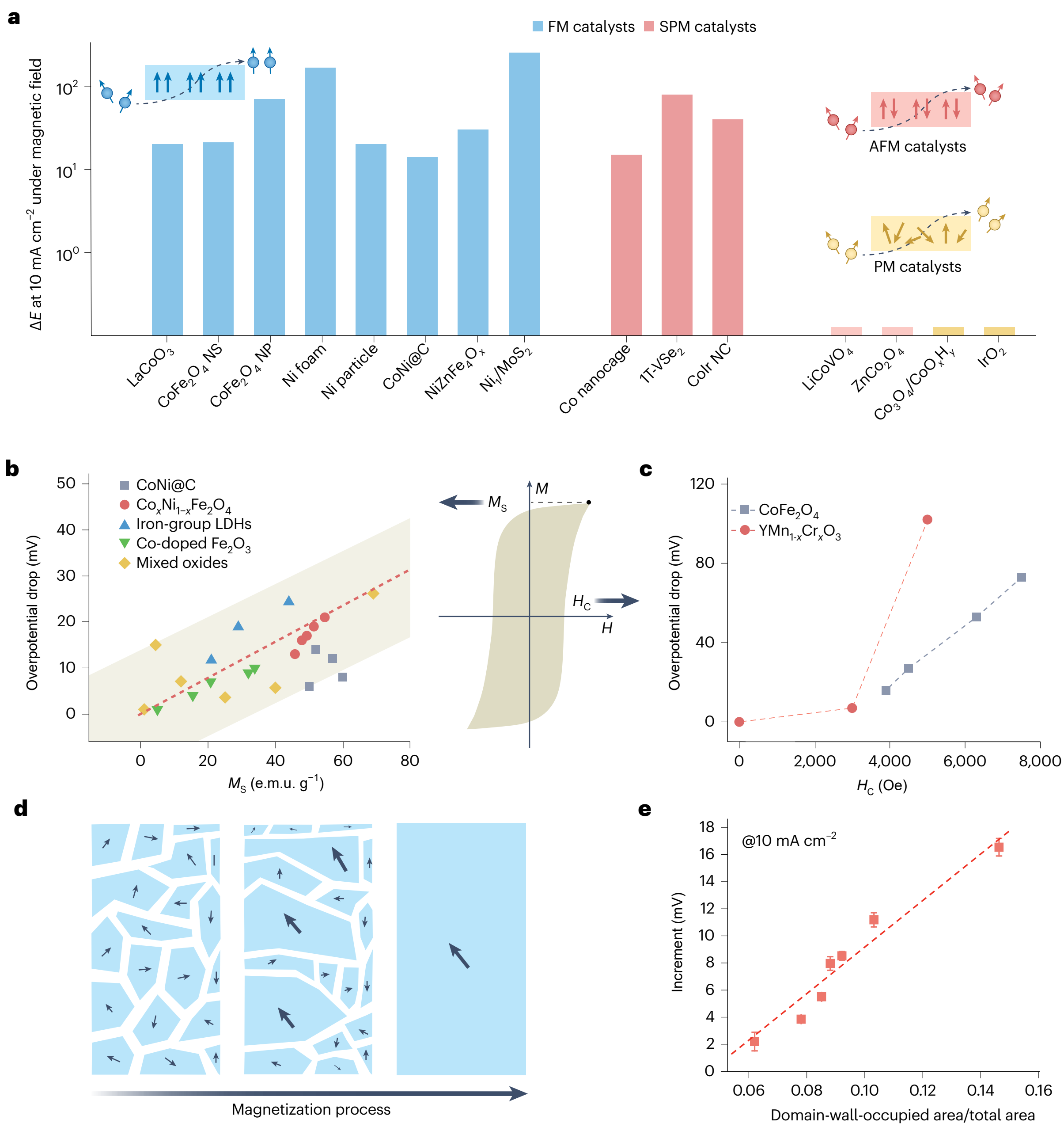

Fig. 4 | Long-range spin ordering for OER enhancement under magnetic fields. a, The reported OER overpotential drop (ΔE, in millivolts) at 10 mA cm^{-2} (normalized to the geometric area of the electrodes) for different catalysts under external magnetic fields, including $LaCoO_3$ under a permanent magnet[90], $CoFe_2O_4$ nanosheet (NS) under a field of 100 mT (ref. 91), $CoFe_2O_4$ nanoparticle (NP) under 1,400 mT (ref. 39), Ni foam under 200 mT (ref. 92), Ni particle under 1,400 mT (ref. 17), CoNi@C under 360 mT (ref. 40), $NiZnFe_4O_x$ under 450 mT (ref. 9), Ni_1/MoS_2 (Ni_1, single-atom Ni) under 502 mT (ref. 68), Co nanocage under 300 mT (ref. 93), 1T-VSe_2 under 800 mT (ref. 94), CoIr nanocluster (NC) under 106 mT (ref. 95), $LiCoVO_4$ and $ZnCo_2O_4$ under 250 mT (ref. 53), Co_3O_4/CoO_xH_y under 500 mT (ref. 12) and IrO_2 under 1,000 mT (refs. 10,96). The insets show the change in spin polarization of electrons by ferromagnetic (FM), antiferromagnetic (AFM) and paramagnetic (PM) materials. **b**,**c**, Dependency of the OER increment on the saturation magnetization (M_S) (**b**) and coercivity (H_C) (**c**), as annotated in the inset magnetic hysteresis loop. The OER enhancement is positively dependent on M_S in CoNi@C (ref. 40), $Co_xNi_{1-x}Fe_2O_4$ (ref. 91), iron-group layered double hydroxides (LDHs)[97], Co-doped Fe_2O_3 (ref. 98) and mixed oxides[9], and it is positively dependent on H_C in $CoFe_2O_4$ (ref. 39) and $YMn_{1-x}Cr_xO_3$ (ref. 99). M, magnetization; H, magnetic field strength; e.m.u., electromagnetic unit. **d**, Schematic illustration of the magnetization process in multi-domain FM materials, which involves the growth of aligned magnetic domains (blue regions) and the disappearance of domain walls (white regions). **e**, The correlation between the OER overpotential drop and the ratio of the surface area occupied by domain walls, where error bars represent the s.d. obtained from three independent measurements[14]. Panels adapted with permission from: **a**, ref. 10, under a Creative Commons licence CC BY 4.0; **e**, ref. 14 under a Creative Commons licence CC BY 4.0.

oxides with the FM double-exchange mechanism (Fig. 6b), indicating the participation of ligand orbitals (lattice oxygen) in the exchange interactions while conserving the spin angular momentum in the electron transfer. Therefore, the FM interatomic QSEIs feature could reduce the electron repulsion, optimize the oxygen intermediate binding energies and enhance the transport of local spin currents[8], resulting in a higher OER activity.

QSEIs provide a theoretical foundation for spin selectivity in reaction kinetics. In OER pathways, such as the adsorbate evolution mechanism (AEM)[70], the lattice-oxygen oxidation mechanism (LOM)[71] and

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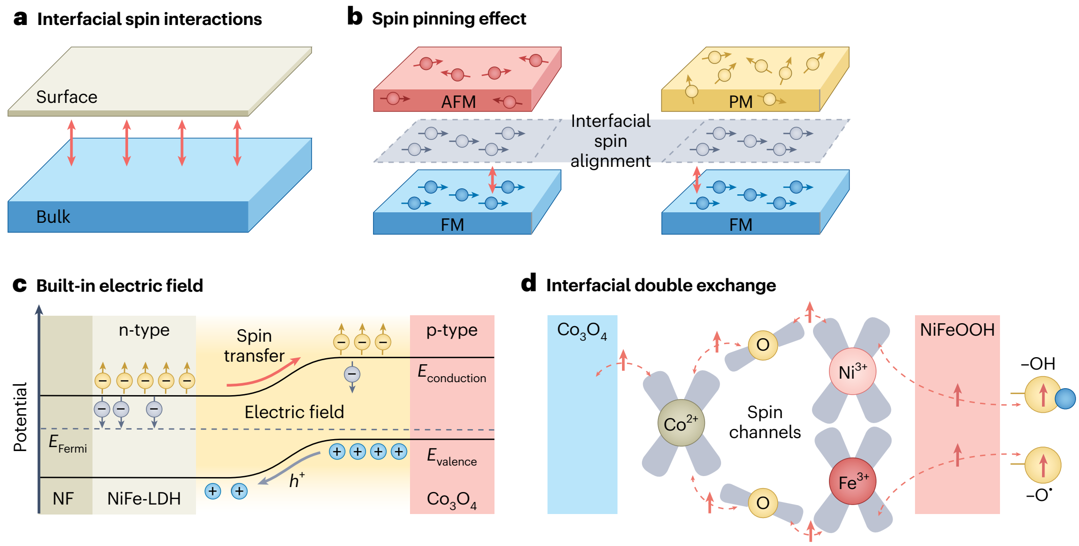

Fig. 5 | Interfacial spin interactions that affect the surface magnetization. **a**, The interfacial spin interactions between bulk and surface catalysts. **b**, Schematic illustration of the spin pinning effect at FM/AFM and FM/PM interfaces. The interfacial spin at the AFM or PM layer could be aligned with the spin in the FM layer. **c**, Schematic energy band diagram of magnetic p–n junction. The inhomogeneous spin split near the depletion region causes spin-polarized electron transport by the built-in electric field[65]. The yellow-coloured and grey-coloured arrows represent spin-up and spin-down electrons, respectively, and h^+ represents the holes that are transported in the valence band. NF, Ni foam; E_{Fermi}, Fermi energy; $E_{conduction}$, conduction band energy; $E_{valence}$, valence band energy. **d**, Example of the double-exchange interaction at the interface. The interaction between Co_{Td}^{2+}–O^{2-}–Ni_{Oh}^{3+} and Co_{Td}^{2+}–O^{2-}–Fe_{Oh}^{3+} creates spin-transfer channels to facilitate the formation of triplet-state oxygen with a lower energy barrier[66]. Panels adapted with permission from: **c**, material and band structure, ref. 63, Wiley; spin transfer mechanism, ref. 65, APS; **d**, ref. 66, American Chemical Society.

the interaction of two metal–oxo entities mechanism (I2M)[72], electron transfer between the active sites and reaction intermediates occurs in multiple steps, making the spin-electron transfer in the whole reaction more complicated than the one-site double-exchange interaction. The spins throughout all OER intermediates are also crucial for understanding the spin-related effects in the whole electron-transfer process. Extending spin polarization from catalysts to reactive intermediates can be mediated by short-range spin-exchange interactions (Fig. 6c)[73]. This phenomenon has been demonstrated in the single-atom Ni_1/MoS_2 electrocatalysts[68]. By substituting the magnetic Ni atoms into the MoS_2, the nearby S atoms showed global ferromagnetism under a magnetic field due to the spin-polarized electron transport between Ni and S. Consequently, the FM-coupled S active sites optimized the spin density in the *OH and *O species, resulting in an aligned spin configuration for both intermediates.

Spin-sensitive OER pathways have been proposed involving an important OER intermediate: the M–O• radical species[6]. Its spin-related role is closely associated with an important experimental finding from the pH-dependent OER enhancement via magnetization[9,12,13]. In the M–O• radical, two electrons in the oxygen p_π orbital and metal d_π orbital, respectively, exhibit parallel spins under the direct exchange interactions (Fig. 6d). When FM coupling is established on metal sites, neighbouring ligand oxygen shows aligned spins[67], which facilitates the radical O–O coupling process to produce the triplet-state O_2 with two parallel spins in $\pi^*_{O_2}$ (refs. 5,6). For one-site pathways such as in the AEM, the spin polarization may favour the selective removal of electrons in σ_{O-H} and σ_{M-O} to form two unpaired electrons in π^*_{O}. Without spin-selective electron removal, the dehydrogenation and the final σ_{M-O} bond cleavage could require a spin flip to reach the preferred spin configuration, which also impedes the kinetics of the OER.

Key factors for observing spin-related OER enhancement

Although the magnetization-enhanced OER should be observed for catalysts with intrinsic FM spin ordering, this enhancement has not yet been universally observed with ferro-/ferrimagnetic catalysts. In some cases, almost no enhancement under a magnetic field on ferro-/ferrimagnetic electrocatalysts has been observed, indicating that additional factors are essential for spin-related effects to be realized in the OER. Here we summarize the key factors for spin-related enhancement under magnetic fields, including the magnetic domain structure, temperature, surface chemistry, pH of the electrolyte and experimental practices.

In a multi-domain structure, domain walls exist in a demagnetized state. As discussed above, the magnetization-induced increase in the OER may originate from the different OER activities in the domain and domain-wall regions. It can thus be inferred that materials with a single-domain nature may not exhibit an OER increase. The transition from a multi-domain structure to a single-domain structure can be achieved by reducing the size of a particle below the critical level equivalent to a single magnetic domain size (Fig. 7a). For example, consider Fe_3O_4 nanoparticles, where the critical size of single-domain Fe_3O_4 is 14–36 nm (ref. 74). Under a magnetic field, 120 nm Fe_3O_4 (multi-domain) is reported to show an enhancement in the OER, whereas the 22 nm Fe_3O_4 (single-domain) remains almost unchanged[11].

These findings also inspired interest in the engineering of single-domain magnetic electrocatalysts to facilitate spin-related kinetics in the OER without using an external magnetic field[59]. The intrinsic activity, as determined by the turnover frequency, of single-domain $CoFe_2O_4$ without magnetization is nearly the same as that of multi-domain $CoFe_2O_4$ with magnetization[59]. This observation is consistent with the fact that the multi-domain $CoFe_2O_4$ transitions to a single-domain state under magnetization. Therefore, for single-domain catalysts, there is no increment observed by applying an external magnetic field, and the maximized promotion via spin polarization occurs naturally on single-domain catalysts. This also indicates that, to observe the enhancement with a magnetic field, the catalysts should have a multi-domain structure.

The intrinsic thermal effect at various temperatures affects the ferromagnetism greatly. Different OER responses to magnetic fields have been reported in FM and PM $La_{0.67}Sr_{0.33}MnO_3$ thin films,[50] in which changing the film thickness changed the Curie temperature (T_C) values

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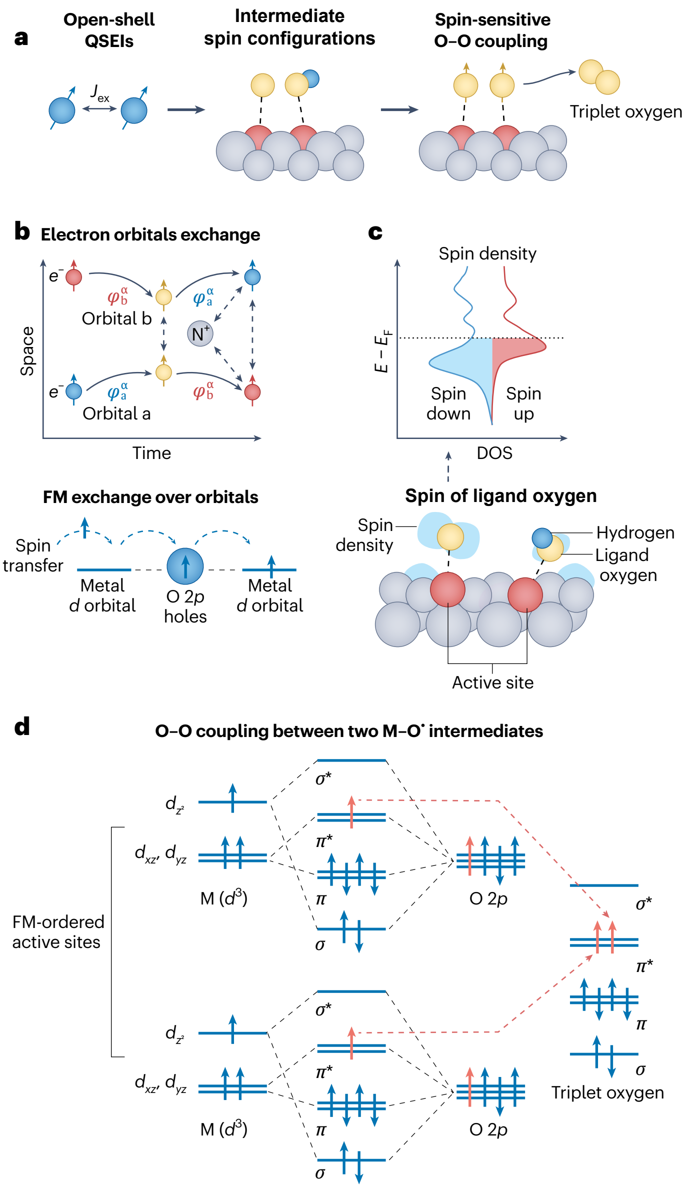

Fig. 6 | Illustration of intermediate spin configuration. **a**, Quantum spin perspective on the generation of triplet oxygen. J_{ex} represents the exchange integral between two spins. **b**, Schematic illustration of the spin-exchange mechanism between two electrons with the same spin represented through a Feynman-type diagram[69] (top) and fundamental exchange interaction over orbitals (bottom). The blue-coloured and red-coloured spin arrows in the upper schematic represent the two electrons with the same spin direction (α) while in different orbitals (φ_a and φ_b). The yellow-coloured spin arrows represent the transition state where two electrons approach each other and exchange their orbitals. The dashed arrows represent the spin electron repulsion and electron–nuclei (e^-–N^+) attraction[12]. **c**, Example of spin polarization in ligand oxygens and the affected density of states (DOS) near the Fermi level (denoted by the dashed line, and where $E - E_F$ denotes the relative energy shift from the Fermi level) for the atomic spin density. **d**, Schematic illustration of O–O coupling between two FM-coupled M–O• intermediates, attributed to spin transfer among molecular orbitals. The aligned spins from the two oxygen species are marked in red, which produces the triplet-state oxygen. Panel **b** adapted with permission from: top, ref. 69, American Chemical Society under a Creative Commons licence CC-BY-NC-ND 4.0; bottom, ref. 67, American Chemical Society.

of the films, such that films with the same chemistry can be either FM or PM at room temperature. The FM film showed an increase in the OER under a magnetic field, whereas the activity in the PM film remained almost unchanged. In addition, enhanced OER activity and facilitated charge transport are observed as the PM film transitions to an FM film below T_C, indicating the thermal disturbance in the long-range magnetic ordering at $T > T_C$. Moreover, the thermal effect has been shown to influence both the enhancement effect via magnetization and the spin pinning effect at temperatures below T_C. For example, a weaker OER enhancement with magnetization has been reported for $CoFe_2O_4$ catalysts as the reaction temperature is increased[10]. This could be attributed to the disturbed magnetic moment in the FM catalyst at elevated temperature, compromising the long-range ferromagnetism, and being less spin-polarized[50]. However, it is worth noting the complex effects of temperature on factors such as OH^- diffusion and surface chemistry. When carrying out Tafel analyses of the magnetic effect at different temperatures, it is crucial to carefully determine whether the temperature change has a non-kinetic influence on the current density[42]. Another example is the breaking of the spin pinning effect observed in the surface-reconstructed $SmCo_5/CoO_xH_y$ catalyst at elevated temperatures[13]. Although the hard magnetic core of $SmCo_5$ has a high T_C of 1,050 K, the spin pinning gradually disappeared when the temperature was increased to 80 °C, as shown by the negligible exchange bias (0.5 Oe). The corresponding spin-enhancement effect of the magnetic field also disappeared at 80 °C. Therefore, there may be a critical temperature (T_{max})—albeit far below the Curie temperature—that is necessary to maintain the spin pinning effect for magnetization-induced enhancement (Fig. 7b), and the OER should be performed below T_{max}.

Stabilized surface chemistry and the thickness of the reconstructed layer are crucial for investigating the spin-related OER under magnetic fields (Fig. 7c). During the alkaline OER, many Ni-, Co- and Fe-based catalysts undergo surface reconstruction to form highly active oxyhydroxide species, leading to increased activity[56]. To avoid the activity contribution from surface reconstruction, sufficient electrochemical conditioning (for example, using cyclic voltammetry) should be carried out for catalysts to stabilize the surface chemistry before conducting electrochemical studies under magnetic fields. In addition to the stable surface chemistry, ideal model catalyst systems for studying the magnetic field-enhanced OER should also benefit from the features of well-defined active sites and controllable local spin structures, which have been demonstrated in some two-dimensional transition metal dichalcogenides[68,75] and single-crystal thin-film materials[50]. Moreover, the thickness of the reconstructed layer could influence the spin pinning effect. For example, Fe_3O_4@$Ni(OH)_2$ core–shell catalysts[11] showed a compromised OER enhancement with magnetization when the shell thickness was increased from 7 to 43 nm, due to weakened spin pinning in the thicker oxyhydroxide layer. For pre-catalysts, an intentional design will be necessary to control the thickness of the reconstructed layer below the effective length of spin pinning.

Enhancement of the OER under magnetic fields depends on the pH of the electrolyte. As summarized in Fig. 7d, two contrasting trends in the magnetic promotion effect on the OER with various electrolyte pH values have been reported, depending on the catalysts. For $Co_{3-x}Fe_xO_4/Co(Fe)O_xH_y$ (ref. 12), $NiZnFeO_x$ (ref. 9) and $SmCo_5/CoO_xH_y$ (ref. 13), the magnetic promotion effect becomes notable as the pH in increased, whereas for Fe_3O_4 and γ-Fe_2O_3, the effect weakens when the pH is increased[22]. These two distinctive trends may be attributed to the different ligand oxygen chemistry in the catalysts at various pH values. Typically, above a specific pH, the absorbed –OH is likely to dehydrogenate to produce M–O• oxyl species, which is an essential intermediate for spin-polarized OER pathways. The pH range for such deprotonation for different catalysts can be variable. Outside the necessary pH range, the OER kinetics become spin-unpolarized and are difficult to be improved by applying a magnetic field, which could be a possible origin of the pH dependency of the magnetic promotion effect. In addition, below a specific pH of the electrolyte, the OH^- transport can be limiting for the OER, which is likely to highlight the OER improvement under the MHD effect.

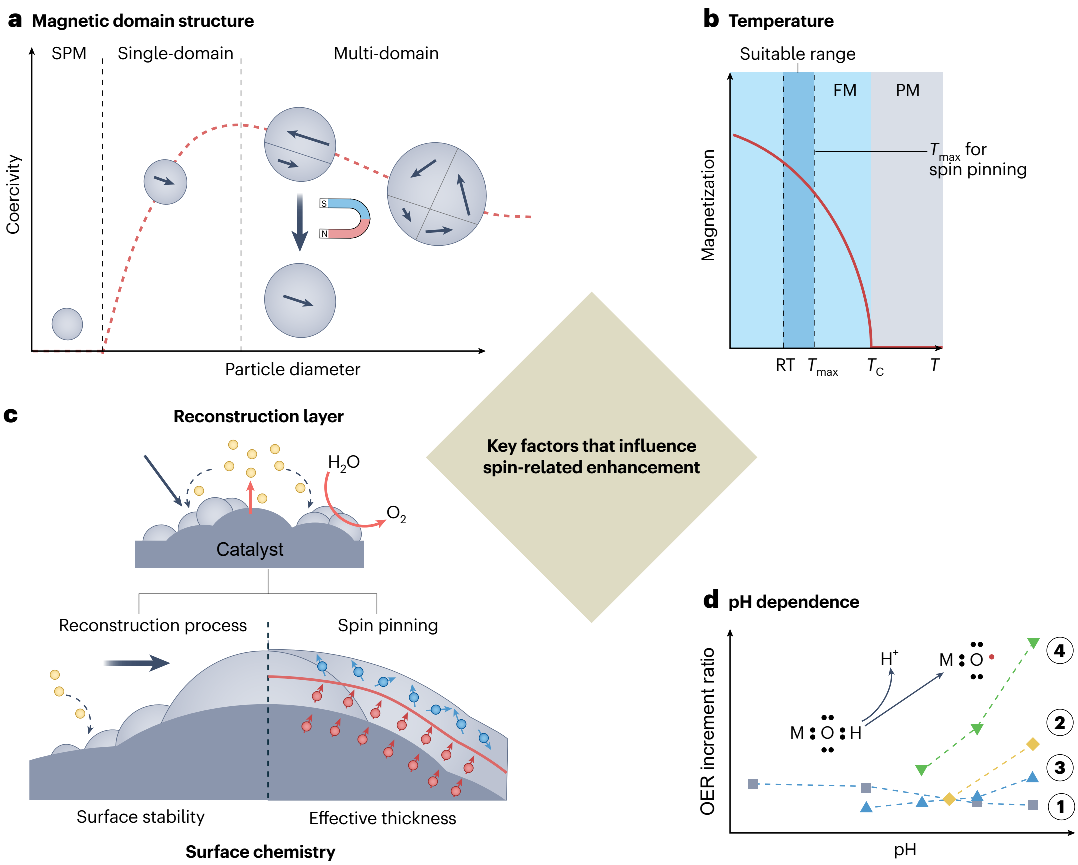

Fig. 7 | Key factors for spin-related OER enhancement. a, Magnetic domain structure of particle materials. Depending on the particle size, the magnetic domain structure varies between the multi-domain, single-domain and SPM state. **b**, Temperature (*T*) effects on the ferromagnetism. A suitable temperature range must be decided according to the effect of thermal disturbance on the FM ordering. RT, room temperature. **c**, Surface chemistry stabilized after electrochemical reconstruction for avoiding activity changes during conditioning and ensuring surface layer within the thickness for effective magnetic effects (for example, spin pinning effect). **d**, Dependence of OER enhancement on the pH of the electrolyte. Data sources: ① Fe_3O_4 (ref. 22), ② $Co_{3-x}Fe_xO_4/Co(Fe)O_xH_y$ (ref. 12), ③ Ni foam[9] and ④ $SmCo_5/CoO_xH_y$ (ref. 13). Inset: deprotonation process for the M–O$^{\bullet}$ oxyl species as a crucial spin-polarized intermediate that relies on strongly alkaline conditions. Panel **a** adapted with permission from ref. 100 under a Creative Commons licence CC BY 4.0.

As discussed above, use of the correct experimental set-up and practices is critical to exclude interference from other effects that can occur under a magnetic field. The recommended set-up in Fig. 2 can minimize effects that include Joule heating, electrode displacement, electromagnetic induction and the Kelvin force. Additional experimental protocols may also be incorporated to examine whether the observed change in OER activity is due to spin-related effects. For example, the applied potential (≤1.60 V versus RHE[9,14,68]) and magnetic field intensity (saturation field strength) should be chosen so as to avoid notable non-kinetic limitations in the OER. If notable MHD effects need to be excluded, mass-transport limitations should be mitigated by, for example, using flow-cell systems[76]. In addition, notable non-kinetic effects (for example, bubble and mass transport) under magnetic fields can influence the ohmic resistance of the electrolyte and affect the potential of the working electrode[36], for which an electrochemical impedance spectroscopy study should be conducted throughout the experiments[42] and *iR* correction performed as necessary. Besides, the magnetic field-enhanced OER performance needs to be compared with both non-magnetic catalysts and demagnetized catalysts to examine the spin-related enhancement[10]. Room-temperature magnetoresistance measurements are also necessary to assess the variation in electric resistance of the electrocatalysts[63].

Opportunities for further advancements

Spin-related OER studies aim to overcome the thermodynamic and kinetic limitations of the OER (Fig. 8a). However, our ability to exploit spin effects to substantially improve the electrochemical performance is currently restricted by our incomplete understanding of them (Fig. 8b).

In this Review we have provided the key factors to help observe the spin-related enhancement (Fig. 8c) and other possible effects of applied magnetization (Fig. 8d), which could unlock more potential for effective spin manipulation. In past studies, spin manipulation without a magnetic field has been achieved using the CISS effect[77]. After understanding the domain wall as one of the origins of OER enhancement under magnetic fields, domain engineering, such as by developing single-domain catalysts, could be an alternative approach for direct spin manipulation[14]. Furthermore, in the OER, spin selectivity in magnetic intermediates can facilitate reaction pathways with low energetic and kinetic barriers. A more fundamental understanding of spin selectivity may also be impactful in controlling the product selectivity in other reactions (for example, nitrate reduction[78], ammonia synthesis[79] and CO_2 reduction[80]), in which identification of the spin polarization of intermediates will be essential.

So far, and despite many contributions to this topic[8,12,14], a few missing pieces still remain from a complete picture. The pursuit of more direct and accurate measurement methods to enhance the reliability of experimental results is ongoing. We put forward that systematic studies of the spin-related OER under magnetic fields in the future could improve our understanding of the OER mechanism and facilitate the development of new magnetic (spin) catalysts. Here we propose some opportunities for further advances.

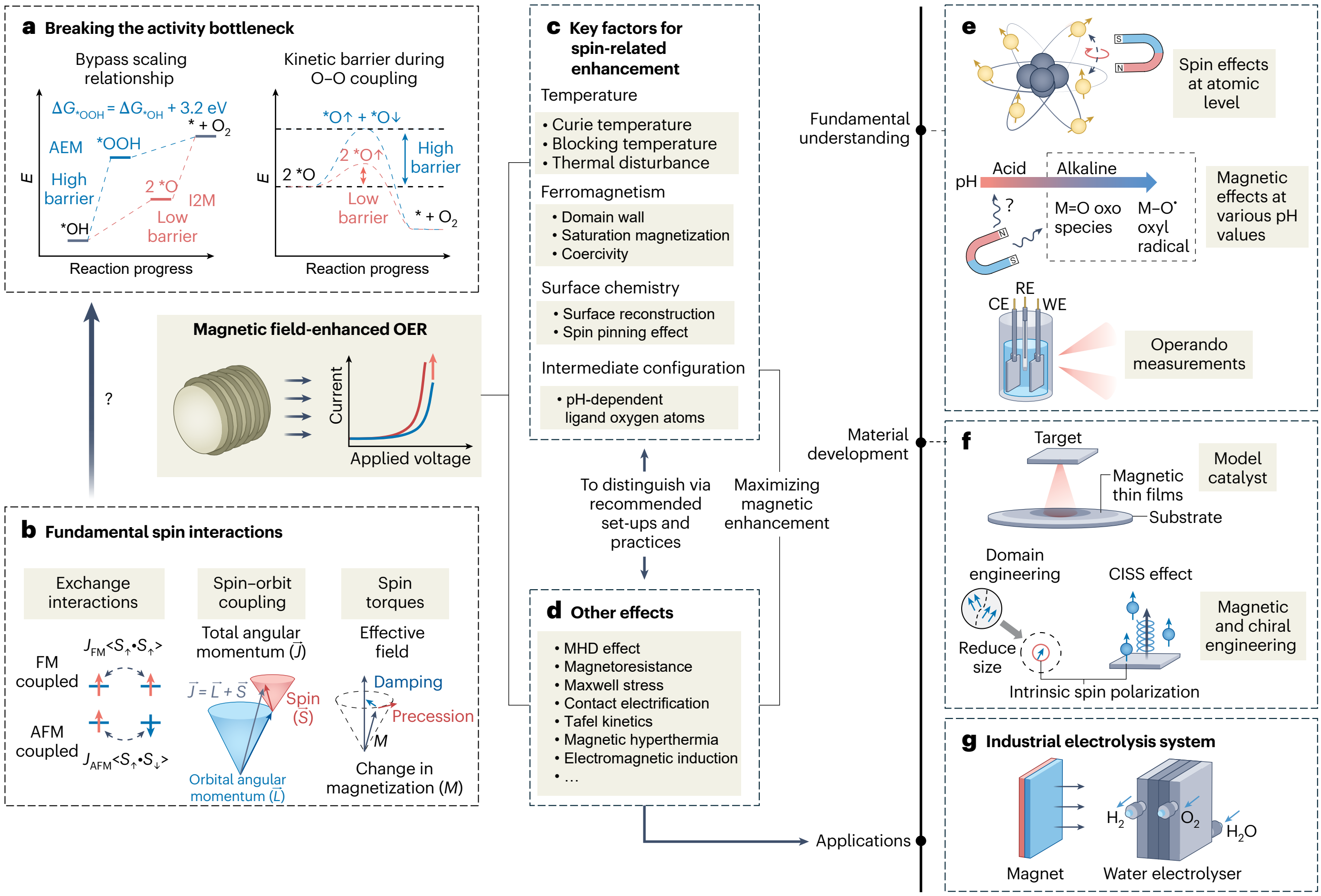

Fig. 8 | Overview and outlook for the OER under magnetic fields. a,b, Studies of the OER under magnetic fields are needed to fill the gap between spin-related energetic and kinetic limitations of the OER (**a**) and the fundamentals of spin interactions (**b**). This gap could potentially be bridged using the magnetic field-enhanced OER. J_{FM}, FM exchange; J_{AFM}, AFM exchange; $S_{\uparrow}$, spin-up momentum; $S_{\downarrow}$, spin-down momentum. **c,d**, Enhancement of the OER with magnetic fields under spin-related effects (**c**) and other effects (**d**). **e–g**, Opportunities for advancing fundamental understanding (**e**), facilitating material development (**f**) and extending our knowledge to industrial applications (**g**). CE, counter electrode; RE, reference electrode; WE, working electrode.

First, mechanisms of the external magnetic field effects at the atomic level need to be explored (Fig. 8e). The commonly applied magnetic field strength for the OER, typically below 1 T (ref. 10), is insufficient to tune the spin angular momentum at ambient temperature[51]. It can only alter the long-range magnetic ordering in ferro-/ferrimagnetic materials, reflecting the average atomic magnetic moment orientation through domain alignment. On the other hand, OER spin-related electron-transfer steps involve atomic or subatomic interactions. In this case, to explore the field-enhanced spin selectivity of the OER further, new findings and mechanisms could probably be revealed using stronger fields or intrinsic spintronics interactions.

Second, the spin selectivity is intrinsic and should be independent of the electrolyte pH. However, enhancement of the OER via magnetization is pH-dependent under alkaline conditions (Fig. 8e). In addition, a potential spin-related effect has also been demonstrated in acidic environments for Mn-doped RuO_2 (ref. 81). This variability may be related to differences in surface chemistry or spin-related reaction pathways under acidic and alkaline conditions[6,12]. A comprehensive understanding of the mechanisms underlying this pH dependence is highly desirable. It is crucial for subsequent electrolyte environment engineering and optimization of the spin configuration of intermediates to fulfil the prerequisites and understand the possible effects on the spin-related kinetics of local pH variation under operando conditions. Therefore, further efforts should be made to investigate the spin-related mechanisms under various pH values using additional experimental and computational methods.

Third, in previous studies of OER enhancement in magnetic electrocatalysis, the magnetic properties of the catalysts were typically evaluated without real-time monitoring under operando conditions. To establish a more accurate correlation to the underlying physics, it is crucial to conduct operando studies of catalyst chemistry under magnetic fields during the actual electrochemical reaction (Fig. 8e). Feasible techniques include operando spectroscopic techniques, such as Raman spectroscopy[48], ultraviolet-visible spectroscopy[82] and X-ray absorption spectroscopy[83], or microscopic methods such as transmission electron microscopy and atomic force microscopy[84]. The magnetic properties of electrocatalysts could be tracked using techniques such as the magneto-optical Kerr effect, magnetic force microscopy, measurement of the anomalous Hall effect[85] and FM resonance[86]. More importantly, additional effort is needed for the integration of these techniques with real electrochemical systems.

Moreover, compared with powder catalysts, thin-film catalysts offer distinct merits that include homogeneous surfaces, clear domain structures, magnetic anisotropy and precise chemical compositions (Fig. 8f). In particular, the domain structure in magnetic thin films can be tailored through modulation of the film thickness[87] or via interfacial strain engineering[88], and it can be observed directly using magnetic force microscopy[89]. A homogeneous surface and chemical composition can further favour the formation of uniform heterostructures via surface reconstruction. These characteristics make thin-film materials advantageous for quantitively correlating the OER increment and change in magnetic properties, making them more suitable for investigating spin-related mechanisms.

Finally, to extend the application of magnetic fields in real electrolysers (Fig. 8g), various factors must be considered. First, at various current densities, it is important to determine whether spin-related or non-spin-related effects are dominating the enhancement. For example, the MHD effect probably dominates the enhancement when mass transport becomes limiting at high current densities. Second, the additional costs of integrating magnetic fields into real electrolysis systems, for example, the capital cost for magnetic components and energy consumption in electric magnets, should be mitigated by device innovation, accompanied by techno-economic analysis. Third, the development of stable magnetic catalysts is a prerequisite for maintaining the long-term benefits of magnetic fields in electrolysers.

Outlook

The application of magnetic fields to enhance the OER has provided new opportunities for exploring spin effects in catalysis, contributing to a deeper understanding that is grounded in catalysis fundamentals. Whereas challenges remain due to the complex effects of magnetic fields, this Review has provided comprehensive guidance for differentiating spin-related effects from other effects, and it has summarized the key factors for observing the OER enhancement caused by spin-related effects. Further studies are desirable to fill the gap between the spin-related fundamentals and catalytic performance, along with extensive endeavours to uncover more details of spin-related mechanisms that span the catalyst bulk, the catalytic interface and intermediates. These efforts will support optimization of the intrinsic spin properties of catalysts and improve their activity. Ultimately, these advancements are expected to influence the design of water electrolysis devices. Beyond OER research, the insights gained from studies of spin-related mechanisms will inspire more considerations of spin effects in other catalytic reactions, with further breakthroughs potentially being made.

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Author contributions

Z.J.X. and T.W. conceived the original idea. A.Y. and T.W. conducted the data collection and data analysis. Y.Z., A.Y. and T.W. designed and prepared all figures. A.Y., T.W. and Z.J.X. wrote the manuscript. S.Z. assisted with editing the language.

Competing interests

The authors declare no competing interests.

Additional information

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